

Lincoln University College

Faculty of Computer Science and Multimedia

RoadSide Assistance System

A PROJECT REPORT

Submitted to

Department of Information Technology

Phoenix College of Management

Maitidevi, Kathmandu

In partial fulfillment of the requirements for the Bachelor in Information
Technology (BIT)

Submitted by

Prajjwol Thapa

BIT 7th Semester

University ID : LC0003001681

2026/03/22

DECLARATION

I hereby declare that the project report entitled "Roadside Assistance System" is my original work, completed under the guidance of Mr. Saishab Bhattarai. This report has not been submitted elsewhere, in full or in part, for the award of any degree, diploma, or other academic recognition.

Signature:

Name: Prajjwol Thapa

Date: 2026/03/21



Lincoln University College
Faculty of Computer Science and Multimedia
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Maitidevi, Kathmandu

Bachelors in Information Technology (BIT)

LETTER OF APPROVAL

The Project report entitled "Roadside Assistance System" submitted by Prajjwol Thapa, in partial fulfillment of the requirements for the Bachelor's degree in Information Technology, has been accepted as a record of work independently carried out by the group of departments.

In our opinion, it is satisfactory in the score and quality of a project for the required degree.

Er. Santosh Dhungana
Head of Department

Mr. Bijay Mishra
Program Coordinator

Mr. Saishab Bhattarai
Internal Examiner

ABSTRACT

The “Roadside Assistance System” is designed to address the growing challenge of vehicle breakdowns and roadside emergencies in Nepal, particularly in rural and highway areas where organized support is limited. By providing a digital platform that connects stranded travelers with nearby available mechanics, the system aims to reduce waiting times, minimize safety risks, and improve accessibility to roadside help. The platform allows users to create service requests, track request status in real-time, and provide feedback, while mechanics can register, manage availability, and accept service requests. The system not only enhances convenience and safety for travelers but also creates income-generating opportunities for local mechanics across different regions, contributing to a more organized and efficient roadside assistance ecosystem.

Keywords: Roadside Assistance, Service Request Management, Digital Platform, Mechanic Network.

ACKNOWLEDGMENT

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Additionally, I extend my gratitude to the **External Advisors** for their encouragement and assistance during the course of this work.

Lastly, I am thankful to all my family and friends who stood by me and helped me persevere to complete this project successfully.

CERTIFICATE

This is to certify that the project report entitled "Roadside Assistance System" submitted by Prajjwol Thapa, University ID: LC0003001681 to Lincoln University College in partial fulfillment of the requirements for the Bachelor in Information Technology, is a record of original work carried out under my supervision and guidance.

Supervisor Name: Saishab Bhattarai

Designation: Project Coordinator

Signature:

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Chapter 1

INTRODUCTION

1.1 Introduction

Roadside emergencies such as tire punctures, battery failure, and minor mechanical issues are common challenges faced by travelers in Nepal, especially in rural and highway areas where organized support is limited. Most people still depend on personal contacts or local workshops, which often leads to delays, inconvenience, and safety risks. Although a few applications like Mobile Garage and Road Sewa exist, their coverage, adoption, and real-time availability features remain limited, leaving many regions underserved.

1.2 Problem Statement

Despite the growing number of vehicles in Nepal, there is no reliable digital system that connects stranded travelers with nearby mechanics or roadside assistance providers. When issues such as flat tires, battery failure, or minor mechanical problems occur, especially in highways, rural areas, or unfamiliar locations, travelers often face significant delays, safety risks, and difficulty in locating help. At the same time, many skilled local mechanics in different regions remain underutilized due to the lack of an organized platform where they can offer their services. The absence of a centralized, real-time assistance system creates inefficiency and inconvenience for both travelers and service providers. Therefore, there is a need for a platform that can link users with nearby available helpers quickly, efficiently, and transparently.

1.3 Project Objectives

- To develop a platform that allows users to request roadside assistance for common vehicle issues.
- To enable helpers to register, manage availability, and receive nearby service requests.
- To manage service requests from creation to completion while supporting local employment.

1.4 Significance of Project

The Roadside Assistance System is a significant project aimed at improving emergency response for travelers facing vehicle breakdowns in Nepal. By connecting stranded users with nearby available mechanics in

real-time, the system reduces waiting times and minimizes safety risks associated with roadside emergencies, particularly in rural and highway areas where organized support is limited.

The platform empowers local mechanics by providing them with a digital channel to offer their services, creating income-generating opportunities and increasing their visibility. For users, it simplifies the process of finding help during stressful situations, enhancing convenience and peace of mind.

Furthermore, the system contributes to the broader transportation ecosystem by promoting organized and efficient roadside assistance services. By bridging the gap between travelers and service providers, this project supports safer roads and fosters a more reliable support network for vehicle owners across different regions of Nepal.

1.5 Limitations of the Project

- **Internet Dependency:** The system requires a stable internet connection to function effectively. Users in remote rural areas or highway regions with poor network coverage may face difficulties creating requests or receiving real-time updates.
- **Limited Geographic Coverage:** The system's effectiveness depends on the availability of registered mechanics in a given area. Regions with fewer registered helpers may experience longer response times or lack of service availability.
- **Limited Service Scope:** The system currently focuses on minor roadside issues such as tire punctures, battery failure, and basic mechanical problems. It does not cover major repairs, towing services, or emergency medical assistance.
- **Multi-language Support:** The application currently supports only the English language, which may limit accessibility for non-English speaking users in Nepal. Future updates could incorporate Nepali and other local languages for broader reach.
- **Payment Integration:** The system does not include built-in payment processing. Transactions between users and mechanics are handled offline, which may lead to pricing inconsistencies or disputes.

1.6 Development Methodology

Scrum Methodology: The "Roadside Assistance System" follows the Scrum methodology, focusing on iterative development with regular sprint planning, daily stand-ups, and sprint reviews. This approach ensures continuous collaboration, quick adaptation to changes, and the delivery of functional increments that meet user needs.

1.7 Organization of the Report

This report is structured into several chapters with each discussing a specific area of the project 'Roadside Assistance System'. The layout is as follows:

Chapter 1: Introduction

Chapter 2: Background Study and Literature Review

Chapter 3: System Analysis

Chapter 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background of the Study

Currently, roadside assistance services in Nepal are limited and largely unorganized. When a vehicle experiences issues such as a flat tire, dead battery, or minor mechanical faults, travelers often rely on personal contacts, local workshops, or bystanders for help. This approach is time-consuming, uncertain, and especially challenging in rural or unfamiliar areas.

In recent years, applications like Mobile Garage and Road Sewa have begun providing digital roadside assistance services in Nepal. These platforms allow users to request help and connect with registered mechanics. While they represent an initial effort to digitize roadside services, their adoption remains limited, and many rural regions are still underserved. Furthermore, these existing applications often have restricted functionality in areas such as real-time availability of mechanics, open registration for local service providers, and coverage of diverse minor vehicle issues.

Overall, the current landscape demonstrates the potential of digital roadside assistance in Nepal but also highlights significant gaps. There is a clear need for a more accessible, user-friendly, and regionally adaptable system that efficiently connects stranded travelers with nearby mechanics. This system should prioritize open registration for local helpers, real-time matching, and broader coverage, particularly in semi-urban and rural areas.

2.2 Literature Review

Roadside vehicle breakdown assistance systems have gained increasing research attention due to the rising number of vehicles and the safety risks associated with unexpected roadside failures. Prior studies primarily focus on leveraging mobile technologies, Global Positioning System (GPS), Internet of Things (IoT), and cloud-based platforms to reduce response time and improve accessibility of roadside services.

Several researchers have explored **real-time vehicle breakdown assistance systems** by integrating GPS and IoT technologies. Pavanakumar *et al.* [1] propose a system that enables stranded users to transmit their real-time

location to service providers using mobile and IoT-enabled communication frameworks. Their study highlights that traditional GSM-based alert systems are slow, localized, and lack efficient resource allocation. Although recent systems improve tracking and response time, limitations remain in terms of real-time mechanic availability and intelligent service distribution.

Aravinthan and Hemalatha [2] present a **web-based breakdown assistance application** aimed at helping users locate mechanics in remote areas. Their literature survey emphasizes that many earlier solutions relied on basic GPS locator mechanisms with limited mechanic databases. A key contribution of their work is the focus on certified and verified mechanics, addressing reliability and security concerns identified in previous systems. However, their findings also indicate challenges in scalability and regional coverage.

Thiagarajan *et al.* [3] examine the limitations of **conventional fixed-range roadside assistance systems**, which are unable to adapt to dynamic roadside conditions. Building on prior research involving real-time mapping and cloud-based authentication, their system integrates hybrid communication models such as cellular and Wi-Fi networks. The reviewed literature suggests that combining mobile platforms with flexible network architectures enhances robustness, particularly in areas with unstable connectivity, though seamless integration of service discovery and allocation remains limited.

A broader comparative study by Suresh Kumar *et al.* [4] reviews multiple roadside assistance platforms, including GPS-based mechanic locators, certified service provider systems, and chatbot-assisted applications. Their analysis reveals that while most systems enhance user convenience through location tracking, they frequently suffer from limited automation, incomplete databases, and dependence on continuous network availability. The authors emphasize the need for dynamic service dispatch and real-time monitoring to improve response efficiency.

Earlier work by Keerthi Priya *et al.* [5] focuses on **mobile-based breakdown assistance finder systems** that connect users with licensed mechanics using GPS services. Their literature review reinforces the importance of trusted mechanic networks and accurate location services. Nevertheless, persistent challenges such as database completeness, mechanic availability, and data security continue to affect system effectiveness.

Chapter 3

System Analysis

3.1 System Analysis

The **Roadside Assistance System** is developed using the **Agile methodology**, following an iterative approach to ensure flexibility and continuous improvement throughout the development lifecycle. The process begins with the requirement analysis phase, where both functional and non-functional requirements are identified, documented, and prioritized based on user needs and system goals. Key requirements such as user registration, service request management, mechanic availability tracking are given high priority to establish the core functionality of the platform.

Based on these requirements, a product backlog is created, listing all features and user stories to be implemented. The backlog serves as a roadmap for the development process, allowing for adjustments as new

insights emerge. The system is then designed and developed in incremental sprints, with each sprint focusing on delivering a specific set of features.

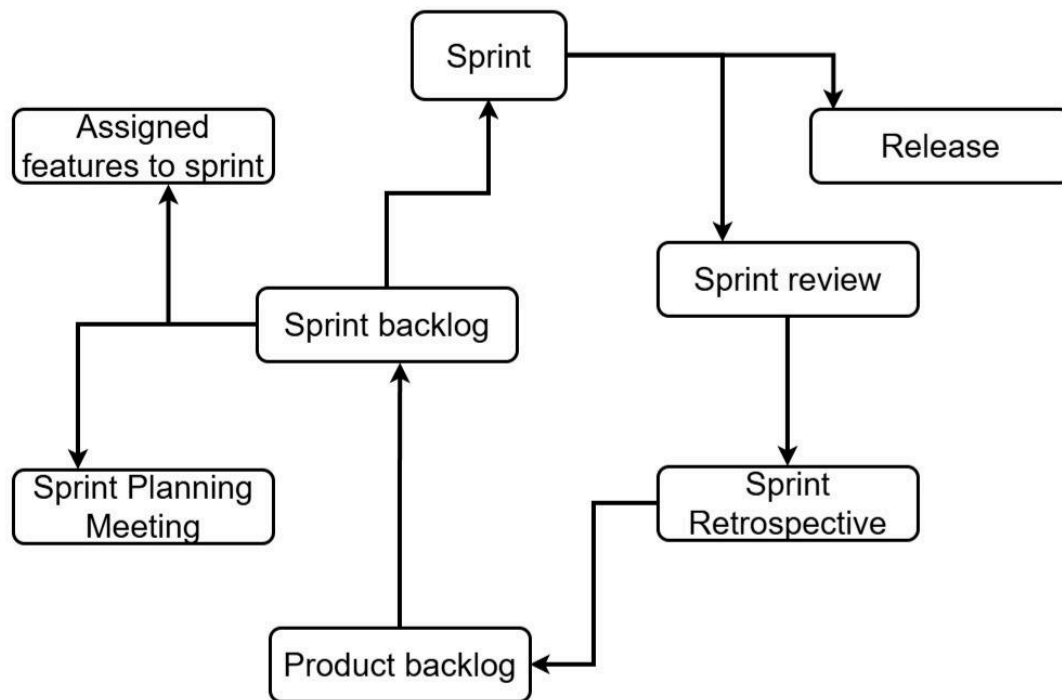


Fig 1: Scrum Methodology

3.1.1 Requirement Analysis

Before commencing the development of the Roadside Assistance System, requirement gathering research was carried out extensively, as the program was intended to address the key challenges faced by travelers and mechanics in Nepal. The requirements were categorized into functional and non-functional requirements to ensure a comprehensive understanding of the system's scope and objectives.

Functional Requirements

- 1. User Registration and Login:** Users must be able to securely register and log in to their accounts to access roadside assistance services, view request history, and manage their profile information.
- 2. Service Request Creation:** Users should be able to create a service request.
- 3. Helper/Mechanic Registration:** Mechanics or helpers must be able to register themselves on the platform by providing their personal details, skills, service area, contact information, and availability status. This enables local service providers to offer their services through the system.
- 4. Availability Management:** Registered helpers should be able to update their availability status at any time. This ensures that only available mechanics are shown to users seeking assistance.

5. **Helper Request Acceptance:** Mechanics should be able to view incoming service requests and choose to accept or decline them.

6. **Feedback and Rating:** After a service is completed, users should be able to rate the helper and provide feedback based on their experience. This helps maintain service quality and builds trust within the system.

Non-functional Requirements

1. **Usability:** The application must be simple and intuitive, allowing users to request assistance with minimal steps. Both users and mechanics should be able to navigate the interface easily without requiring extensive training.

2. **Reliability:** The system should maintain consistent and accurate data, ensuring that service requests, user information, and helper availability are always correctly reflected.

3. **Performance:** The system should respond quickly to service requests.

4. **Availability:** The platform should be accessible whenever users need assistance, with minimal downtime.

5. **Compatibility:** The system should be compatible with modern web browsers and function smoothly.

3.1.2 Feasibility Analysis

To ensure the successful development and implementation of the Roadside Assistance System, a thorough feasibility study was conducted from technical, operational, economic, and scheduling perspectives.

Technical Feasibility

The project utilizes Next.js for frontend and backend development, providing server-side rendering and built-in API routes that streamline development. SQLite serves as the database, offering a lightweight, file-based solution requiring minimal configuration. All tools are open-source, well-documented, and supported by strong developer communities.

Operational Feasibility

The system features an intuitive interface allowing travelers to request assistance in minimal steps, while mechanics can easily register, manage availability, and accept requests. As a web-based platform, it requires no installation and works across smartphones, tablets, and desktops. Users with varying technical skills can navigate the system effortlessly, ensuring high operational feasibility.

Economic Feasibility

Development costs are minimized through the use of free, open-source technologies including Next.js, SQLite, and Node.js. No paid licenses or specialized hardware are required. Hosting can be done on low-cost shared servers or free cloud tiers such as Vercel. This makes the project economically feasible with minimal financial investment.

Schedule Feasibility

The project follows a structured development plan with clearly defined phases and milestones. Next.js simplifies development by combining frontend and backend in a single framework, while SQLite eliminates complex database configuration. The modular architecture enables parallel development of core features, reducing the risk of delays and ensuring timely completion.

3.1.3 System Modeling

3.1.3.1 Use Case Diagram:

A Use Case Diagram is a visual representation of the interactions between users (or actors) and a system. It captures the functional requirements of the system by illustrating the various ways users interact with the platform.

Actors

The Roadside Assistance System involves two primary actors:

1. **User** – A traveler experiencing vehicle issues who seeks roadside assistance. The user interacts with the system to request help, view available mechanics.
2. **Helper** – A mechanic or service provider who offers roadside assistance. The helper interacts with the system to manage their availability, accept or reject service requests, and update service status.

Use Cases

The following use cases have been identified for each actor:

User

- Register Account and Login.
- Create Service Request and View Available Helpers
- Send Service Request

Helper

- Register Account and Login.
- Manage Availability and View Service Requests
- Accept/Reject Service Request

System

- Authenticate Users & Helpers
- Manage Request Status
- Send Notifications

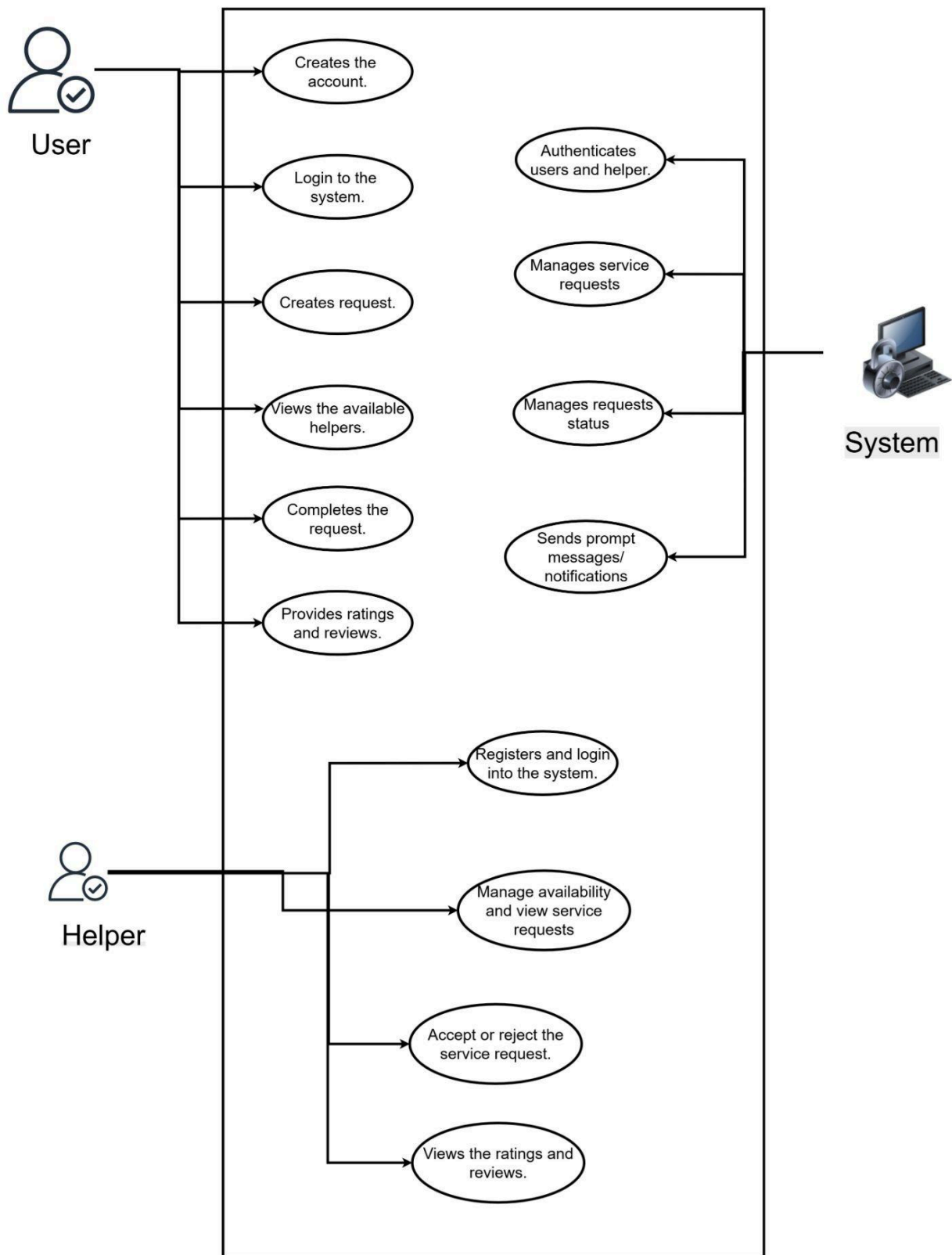


Fig 2: Use Case Diagram of RoadSide Assistance System

3.1.3.2 Class Diagram:

A Class Diagram shows the system's classes, their attributes, methods, and the relationships between them. It provides an overview of how the system is organized and how objects interact.

1. User

- Attributes: userID, name, phoneNumber, email, location, userType
- Methods: requestService(), selectHelper(), cancelRequest(), provideRating()

2. Vehicle

- Attributes: vehicleID, vehicleType, vehicleNumber, problemDescription
- Methods: updateProblem()

3. ServiceRequest

- Attributes: requestID, requestTime, status, serviceType, userLocation
- Methods: createRequest(), updateStatus(), assignHelper()

4. Helper

- Attributes: helperID, name, phoneNumber, email, serviceType, availabilityStatus, averageRating
- Methods: viewRequest(), acceptRequest(), declineRequest(), completeService()

5. RatingReview

- Attributes: reviewID, rating, comment, reviewDate
- Methods: submitReview(), editReview()

Relationships

- A User can create multiple ServiceRequests (one-to-many)
- A ServiceRequest is assigned to one Helper (many-to-one)
- A User owns one Vehicle (one-to-one)
- A User can submit multiple RatingReviews for different services (one-to-many)
- A Helper can receive multiple RatingReviews (one-to-many)

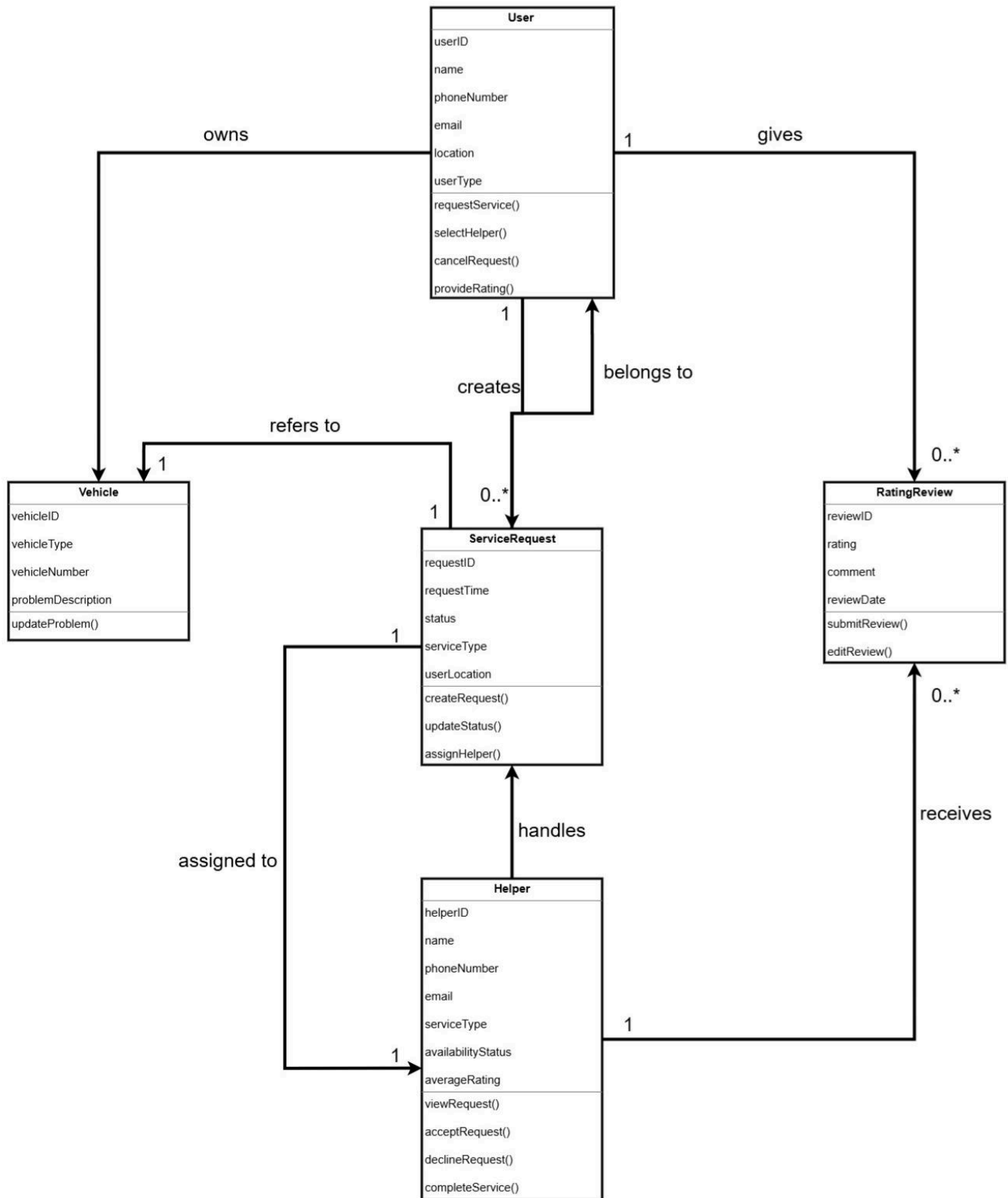


Fig 3: Class Diagram of RoadSide Assistance System

3.1.4 Dynamic modeling using Sequence Diagram

A Sequence Diagram visually represents how objects or components in the system interact over time. It helps to understand the flow of processes and the order in which operations occur.

Actors and Components

- Driver/User – The stranded traveler requesting assistance
- Application – Frontend interface
- Backend API – Handles business logic and request processing
- Database – Stores user, helper, and request data
- Notification Service – Manages real-time notifications
- Service Provider (Helper) – Mechanic providing assistance
- Location Service – Manages location tracking and matching

Interaction Flow

1. User requests assistance with their location
2. Application sends assistance request and location coordinates to Backend API
3. Backend API validates the request and stores it in Database
4. Database confirms storage with request ID (e.g., REQ-001)
5. Backend API queries Location Service for nearby available helpers
6. Location Service returns list of available providers
7. Backend API sends notification via Notification Service to nearby helpers
8. Notification Service delivers request to Service Provider
9. Helper accepts the request
10. Acceptance is sent back through Backend API
11. Database updates request status to "Accepted"
12. Notification Service notifies User of acceptance
13. User receives confirmation with provider details and ETA

Alternative Flows

Helper Rejects Request:

- Helper marks request as rejected
- System updates status to "Rejected"
- Notification Service notifies next available helper

User Cancels Request:

- User cancels the request
- Database updates status to "Cancelled"
- Notification Service notifies the helper (if already assigned)

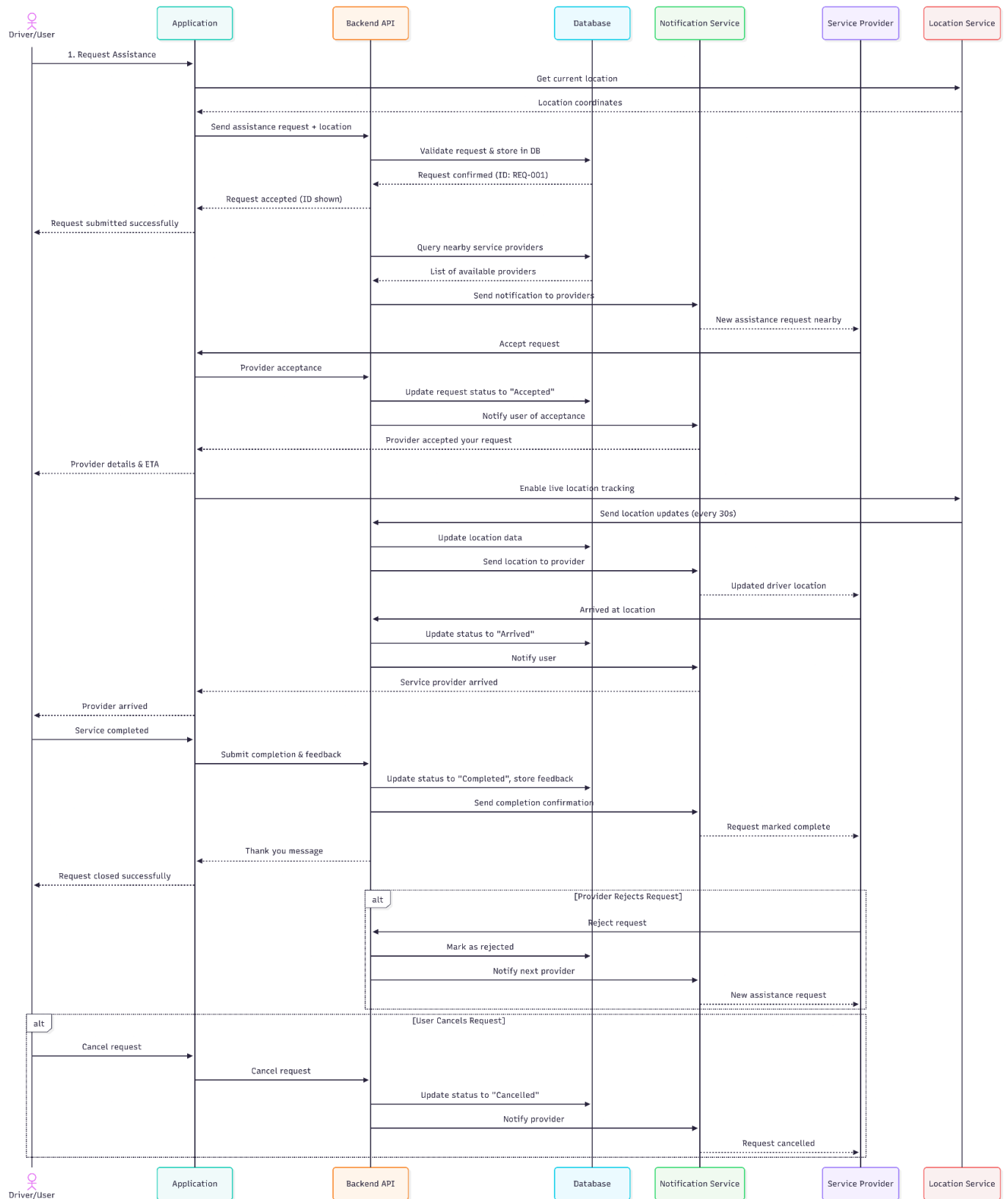


Fig 4: Sequence Diagram of Roadside Assistance System

3.1.5 Process modeling using Process Flow Diagram

A Process Flow Diagram visually represents the sequential steps and decision points within a system. It helps illustrate the logical flow of operations, showing how users and helpers navigate through the system to complete a service request.

Process Flow

The following process flow describes the user journey from registration to service completion:

1. **Start** – The process begins when a user accesses the system.
2. **Account Check** – The system determines whether the user has an existing account.
 - No – User registers a new account by providing required details.
 - Yes – User proceeds to login.
3. **Login** – User authenticates to access the system dashboard.
4. **Navigate to Dashboard** – User views the main interface with available options.
5. **Create Request** – User creates a service request by filling in required fields such as issue type, location, and vehicle details.
6. **Request Visibility** – The created request appears on the helper side dashboard, where nearby available helpers can view it.
7. **Helper Decision** – Helper reviews the request and makes a decision:
 - Accepts Request – Helper accepts and proceeds to the user's location.
 - Declines Request – Request remains available for other helpers.
8. **Vehicle Repair** – Helper reaches the location and performs the required roadside repair.
9. **Mark Completed** – Both helper and user mark the request as completed after service is finished.
10. **Review and Rating** – User provides feedback and rating based on their service experience.
11. **End** – The process concludes.

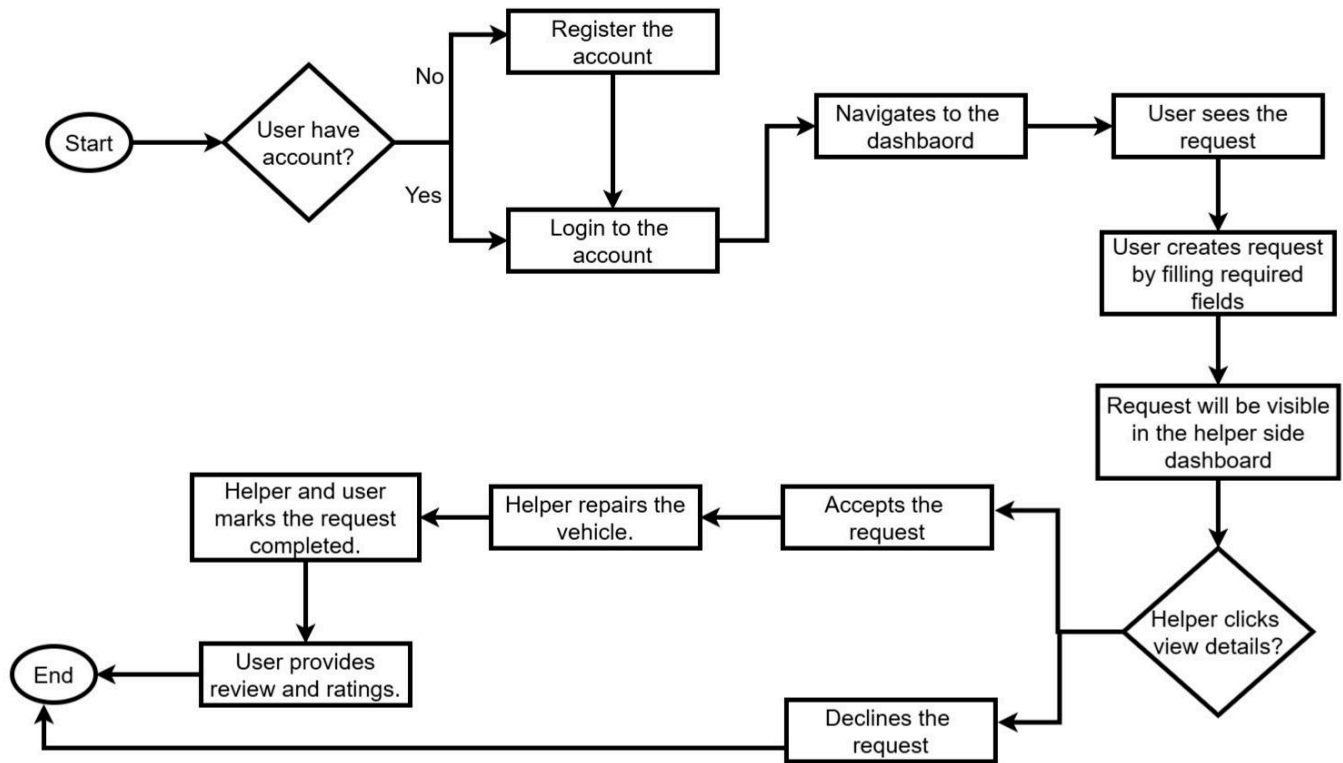


Fig 5: Process Flow Diagram of Roadside Assistance System

References

[1] P. Pavanakumar, J. Jayakaruna, B. R. Ganesh, R. Ramkrishna, and K. Y. Nagababu,

“On Road Vehicle Breakdown Assistance,” International Research Journal on Advanced Engineering Hub (IRJAEH), vol. 3, no. 9, pp. 3724–3729, Sep. 2025, doi: 10.47392/IRJAEH.2025.0541.

[2] C. S. Aravinthan and M. Hemalatha,

“On Road Vehicle Breakdown Help Assistance Web Application,” International Journal of Scientific Research in Science and Technology (IJSRST), Mar.–Apr. 2024, doi: 10.32628/IJSRST52411274.

[3] G. Thiagarajan, B. Santhosam, S. R. David, K. A. Deenadhayal, and A. J. Nathan,

“Vehicle Breakdown Assistance System for On-Road,” International Research Journal on Advanced Engineering Hub (IRJAEH), vol. 3, no. 4, pp. 1781–1785, Apr. 2025, doi: 10.47392/IRJAEH.2025.0257.

[4] R. G. Suresh Kumar, K. R., S. J., G. G., and M. Shanmugadasan,

“Roadside Assistance for Vehicle Breakdown,” International Journal for Research in Applied Science and Engineering Technology (IJRASET), May 2025, doi: 10.22214/ijraset.2025.70634.

[5] K. Keerthi Priya, G. Hari Prasad, G. Sai Prakash, and V. Lingamaiah,

“On Road Vehicle Breakdown Assistance Finder,” International Journal of Scientific Research in Science and Technology, Mar. 2023, doi: 10.32628/IJSRST52310219.